Lesson 3 Teacher's Key

GDevelop

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution

Two events: one to spawn, one that always moves every existing Bullet instance. The Destroy outside screen behavior (added once, on the Bullet object) handles cleanup with zero events:

```
Condition: Key pressed [Space]
Action: Create object Bullet at (Player.X(), Player.Y())

(always, no condition)
Action: Bullet.Y = Bullet.Y() - 400 * TimeDelta()

(Bullet object has the Destroy outside screen behavior)
```

Level 2 - Complete Worked Solution

Unlike the code-based engines, GDevelop's "Key pressed" condition is level-triggered (true on every frame the key is held), so a cooldown is required here, not optional, to avoid flooding the screen with bullets:

```
At the beginning of the scene:
Action: Start (or reset) a scene timer "FireCooldown"

Every frame:
Condition: Key pressed [Space]
Condition: Timer value "FireCooldown" >= 0.2 seconds
Action: Create object Bullet at (Player.X(), Player.Y())
Action: Start (or reset) a scene timer "FireCooldown"
```

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

Sound: play on fire

Add a "Play a sound" action (using sfx_shoot.wav) to the same event that creates a Bullet:

```
Condition: Key pressed [Space]
Condition: Timer value "FireCooldown" >= 0.2 seconds
Action: Create object Bullet at (Player.X(), Player.Y())
Action: Start (or reset) a scene timer "FireCooldown"
Action: Play a sound (sfx_shoot.wav)
```

Visual: give the bullet its own look

"Tint color" and "Change the scale of an object" applied when the Bullet is created, so every spawned copy looks distinct - plus Bullet's own per-instance object timer (not the scene's shared "FireCooldown" timer) counting down in a second, always-running event so the flash actually fades back to normal instead of staying permanent:

```
Condition: Key pressed [Space]
Condition: Timer value "FireCooldown" >= 0.2 seconds
Action: Create object Bullet at (Player.X(), Player.Y())
Action: Tint color of Bullet = Yellow
Action: Change the scale of Bullet to 1.3
Action: Start (or reset) the object timer "Flash" of Bullet

Every frame (separate event, no Space condition):
Condition: Bullet - Timer value "Flash" >= 0.1 seconds
Action: Tint color of Bullet = White
Action: Change the scale of Bullet to 1
```

Structure: group Bullet-specific events together

Keep every Bullet-related event - the fire trigger, the always-on movement, and anything from the ideas above - inside one labeled event group (e.g. "Shooting"), separate from the Player's own movement events carried over from Lesson 2. GDevelop's event sheet has no hard file boundary between objects the way separate scripts/objects do in other engines, so this labeled grouping is what keeps the two mechanics from blurring together.

Complete Level 3 solution (all 3 goals combined)

Everything above, organized as one event group named "Shooting":

```
At the beginning of the scene:
Action: Start (or reset) a scene timer "FireCooldown"

Event group: "Shooting"

  Condition: Key pressed [Space]
  Condition: Timer value "FireCooldown" >= 0.2 seconds
  Action: Create object Bullet at (Player.X(), Player.Y())
  Action: Start (or reset) a scene timer "FireCooldown"
  Action: Play a sound (sfx_shoot.wav)
  Action: Tint color of Bullet = Yellow
  Action: Change the scale of Bullet to 1.3
  Action: Start (or reset) the object timer "Flash" of Bullet

  (always, no condition)
  Action: Bullet.Y = Bullet.Y() - 400 * TimeDelta()

  Condition: Bullet - Timer value "Flash" >= 0.1 seconds
  Action: Tint color of Bullet = White
  Action: Change the scale of Bullet to 1

(Bullet object has the Destroy outside screen behavior - handles cleanup
with zero events)
```